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* MNIST Handwritten Digit Classification

Introduction

Handwritten digit classification is an important application of machine learning and deep learning. The purpose of this project is to develop a system that can identify handwritten digit automatically.

In this project the MNIST dataset is used. MNIST contains handwritten images of digits from 0 to 9. Each image is a grayscale image of size 28×28 pixels.

The dataset contains 60,000 training images and 10,000 testing images. Training images are used to teach the model while testing images are used to check the performance of the trained model.

MNIST Dataset

MNIST is a labeled dataset containing ten classes:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Each image consists of 784 pixels because:

$$28 \times 28 = 784$$

The pixel values range from 0 to 255. These values are normalized between 0 and 1 before training.

Data preprocessing

Data preprocessing is performed before training the model. The pixel values are divided by 255 so that they fall between 0 and 1.

This normalization makes the input data easier for the neural to process and helps improve the training process.

The images are then provided to the neural network for learning.

→ Neural Network Architecture

A simple neural network is used for handwritten digit classification.

The architecture consists of an input layer, a hidden layer, and an output layer.

Input layer

The input consists of a 28×28 pixel image. Each image contains 784 pixel values.

~~Assignment~~

Flatten layer :-

The image converted from a two-dimensional 28×28 matrix into a one-dimensional array of 784 values.

Hidden layer :- The hidden layer contains 128 neurons. The ReLU activation function is used. ReLU helps the neural network learn complex and non-linear patterns from the images.

Output layer :- Output layer contains 10 neurons representing the ten possible digits from 0 to 9.

The Softmax activation function is used to calculate the probability for each digit. The digit having the highest probability is selected as the predicted result.

Model training :- The neural network is trained using the training images and their labels.

During training the model compares its prediction with the correct answer. It then adjusts its weights to reduce the error.

The model is trained for 10 epochs with a batch size of 32.

During training, accuracy, validation accuracy, training loss, and validation loss are recorded.

Model Evaluation and Prediction

After training, the model is evaluated using the 10,000 test images.

The main performance measurement is accuracy. Accuracy represents the percentage of correctly classified images.

For example, if the model correctly predicts 9,700 out of 10,000 images, the accuracy is 97%.

A simple neural network for MNIST can normally achieve a high accuracy, although the exact result depends on the training process.

Accuracy and Loss

Training accuracy and validation accuracy are used to understand how well the model is learning.

As training progresses, accuracy generally increases. At the same time, the loss generally decreases.

The accuracy graph shows the improvement of the model during training, while the loss graph shows the reduction in prediction error.

Experiment and Result

Experiment : changing number of neurons

An experiment is performed by changing the number of neurons in the hidden layer.

The original model uses 128 neurons. In the experiment model, the number of neurons is increased to 256.

model	neurons	Test accuracy
model - 1	128	—
model - 2	256	—

The accuracy values should be filled using the actual results obtained during execution.

Conclusion :

The project demonstrates the use of a neural network for handwritten digit classification. It provides an understanding of data preprocessing, neural-network architecture, activation functions, training, testing, accuracy, loss, and prediction.

The experiment with different number of neurons shows how changing the structure of a neural network can affect its learning process and performances.